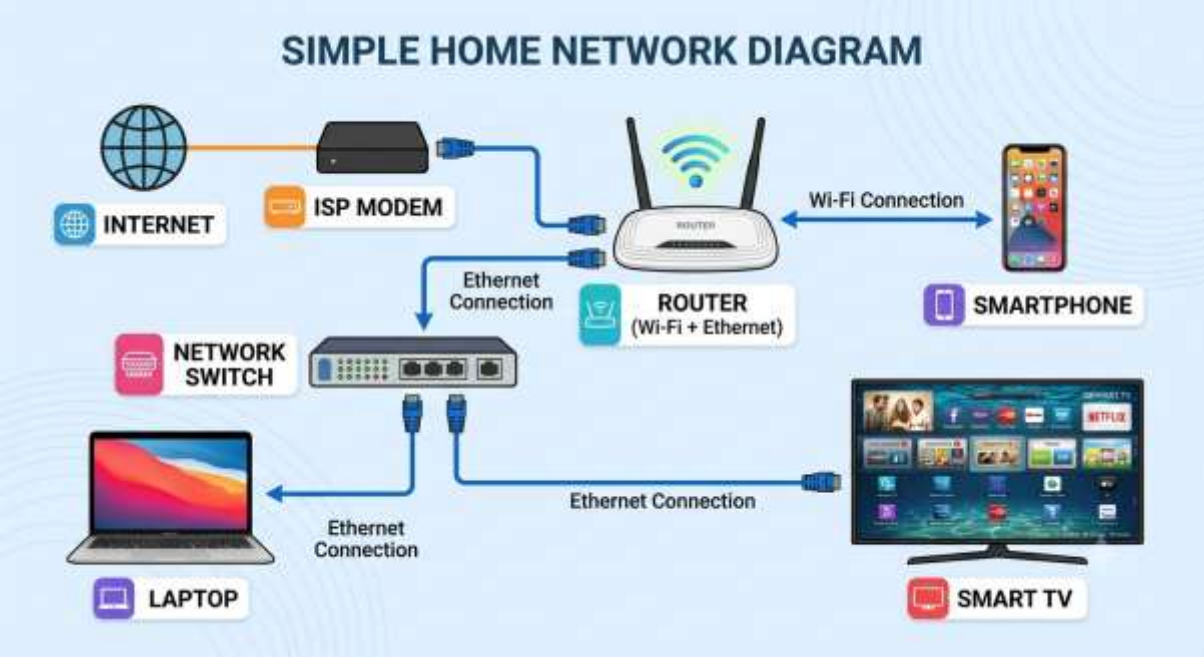
1.

Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.\

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2.

Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

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Comparison of Network Devices and Their Real-World Uses

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and forwards data packets between them.	When I watch YouTube videos on my phone, the router forwards data between my home network and the internet.
Switch	Connects multiple devices within a LAN and forwards data to the intended device.	In a college computer lab, a switch connects multiple computers so they can access shared files and network services.
Hub	Connects devices in a LAN and broadcasts incoming data to all connected ports.	In an older computer lab, a hub could connect multiple computers for basic file sharing.

Access Point (AP)	Provides Wi-Fi connectivity to wireless devices.	A college access point allows students to connect their phones and laptops to Wi-Fi to use Instagram or attend online classes.

3.

Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

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Data Journey Scenario: Ordering Food on Zomato

Imagine you are relaxing on your couch and open the **Zomato** app on your smartphone to order your favorite dinner. Here is how your data travels across the network devices:

- **Smartphone to Access Point / Wireless Router:** When you tap to place your order, your phone sends out radio frequency packets containing your order request. These wireless signals are captured by the **Access Point** (or the built-in Wi-Fi access point feature of your home wireless router).
- **Router Processing:** The **Router** receives this wireless data, translates it, and inspects the destination IP address. It determines that the destination is outside your home network and routes the packet toward your Internet Service Provider (ISP) gateway.
- **Switch Role (Optional/Local Routing):** If your home setup routes certain wired local traffic or uses an intermediate **Switch** to link multiple wired access points or devices before hitting the main router, the switch ensures local frame forwarding efficiency.
- **To the Internet:** Finally, the packet exits your router through the ISP modem out to the public internet, traveling to Zomato's cloud servers to confirm your food order and send back the receipt!

4.

Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

ANS:-

Connecting a Distant Gaming PC via Ethernet

If your gaming PC only has an Ethernet port and is located far from your primary Wi-Fi router, you can bridge the distance using one of these two effective devices:

Option 1: Powerline Adapter Kit

- **How it works:** A Powerline kit comes in a pair of adapters. You plug the first adapter into a wall outlet near your router and connect it to the router using a short Ethernet cable. You plug the second adapter into an electrical outlet near your gaming PC and connect it to your PC's Ethernet port. The adapters use your home's existing electrical wiring to transmit high-speed digital network data between the two rooms, giving you a stable wired connection without running a long physical cable across the house.

Option 2: Wireless Range Extender (with Ethernet Port)

- **How it works:** You place a Wi-Fi range extender in a wall outlet midway between your router and your gaming PC, ensuring it remains within strong wireless range of the router. The extender connects wirelessly to your router's Wi-Fi network. Because many modern range extenders feature an Ethernet port on the bottom, you can run a standard Ethernet cable from the extender's port directly into your gaming PC's Ethernet port. The extender acts as a bridge, converting the wireless signal from your router into a wired Ethernet connection for your PC.